

# AI BlogNest API

## UAT Test Execution Report

|                |                                       |
|----------------|---------------------------------------|
| Date           | 26 September 2026                     |
| Team ID        | SWTID-2026-6999                       |
| Project Name   | AI BlogNest API                       |
| Maximum Marks  | 20 Marks                              |
| Testing Period | 24 September 2026 - 26 September 2026 |

## 1. Project Overview

|                 |                                        |
|-----------------|----------------------------------------|
| Project Name    | AI BlogNest API                        |
| Project Version | 1.0                                    |
| Testing Period  | 24 September 2026 to 26 September 2026 |

**Project Description:** AI BlogNest API is a backend REST API for blog content management and AI-assisted content creation. The system provides user authentication, role-based access control, blog management, categories, comments, Google Gemini-based blog generation and summarization, keyword search using MongoDB Atlas Search, and semantic search using MongoDB Atlas Vector Search.

## 2. Testing Scope

The following functionalities are tested:

- User Registration
- User Login
- JWT Authentication
- Role-Based Access Control
- User Role Management
- Blog Creation
- Blog Retrieval
- Blog Update
- Blog Deletion
- Blog Status Management
- Category Management
- Comment Creation
- Comment Retrieval
- Comment Moderation
- AI Blog Generation
- AI Blog Summarization
- MongoDB Atlas Search
- MongoDB Atlas Vector Search
- Semantic Search
- Input Validation
- Unauthorized Access Handling
- API Rate Limiting
- Input Sanitization (XSS Prevention)
- Content Discovery by Category

The UAT scope identifies the features, functionalities, user stories, and requirements being tested, including three additional scenarios covering rate limiting, input sanitization, and category-based content discovery.

## 3. Testing Environment

|                      |                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------|
| Application Type     | Backend REST API                                                                                 |
| Testing Tool         | Postman                                                                                          |
| Backend              | Node.js + Express.js                                                                             |
| Database             | MongoDB Atlas                                                                                    |
| AI Service           | Google Gemini API                                                                                |
| Search Services      | MongoDB Atlas Search and MongoDB Atlas Vector Search                                             |
| Application Location | http://localhost:8000                                                                            |
| Credentials          | Test user accounts with appropriate roles are used for authentication and authorization testing. |

## 4. Test Cases

The following 25 test cases cover both positive and negative scenarios, as required by the UAT template. All test cases were executed during the testing period and passed.

| TC ID  | Test Scenario                                    | Test Steps                                                                                                                      | Expected Result                                                                                  | Actual Result                                                                     | Pass/Fail |
|--------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------|
| TC-001 | User Registration with valid details             | 1. Send registration request.<br>2. Enter valid name, email and password.<br>3. Submit request.                                 | User account should be created successfully.                                                     | Account created successfully; 201 response returned.                              | Pass      |
| TC-002 | Registration with duplicate email                | 1. Use an already registered email.<br>2. Submit registration.                                                                  | System should reject duplicate registration.                                                     | Request rejected with duplicate-email error.                                      | Pass      |
| TC-003 | User Login with valid credentials                | 1. Send login request.<br>2. Enter valid email and password.                                                                    | Login should succeed and JWT token should be returned.                                           | Login succeeded; JWT token returned.                                              | Pass      |
| TC-004 | Login with invalid password                      | 1. Enter valid email.<br>2. Enter incorrect password.<br>3. Submit.                                                             | Login should be rejected with an authentication error.                                           | Login rejected with authentication error.                                         | Pass      |
| TC-005 | Access protected API without JWT                 | 1. Open a protected endpoint.<br>2. Do not provide Authorization header.                                                        | Request should be rejected as unauthorized.                                                      | Request rejected with 401 Unauthorized.                                           | Pass      |
| TC-006 | Author creates a blog                            | 1. Login as Author.<br>2. Use JWT token.<br>3. Send Create Blog request.                                                        | Blog should be created successfully.                                                             | Blog created successfully; 201 response returned.                                 | Pass      |
| TC-007 | Reader attempts to create blog                   | 1. Login as Reader.<br>2. Send Create Blog request.                                                                             | Request should be rejected with authorization error.                                             | Request rejected with 403 Forbidden.                                              | Pass      |
| TC-008 | Retrieve published blogs                         | 1. Send GET /api/blogs.<br>2. Check response.                                                                                   | Published blogs should be returned successfully.                                                 | Published blogs returned successfully.                                            | Pass      |
| TC-009 | Author updates own blog                          | 1. Login as Author.<br>2. Select own blog.<br>3. Send update request.                                                           | Blog should be updated successfully.                                                             | Blog updated successfully.                                                        | Pass      |
| TC-010 | Author updates another author's blog             | 1. Login as Author 2.<br>2. Attempt to update Author 1's blog.                                                                  | Access should be denied.                                                                         | Access denied with 403 Forbidden.                                                 | Pass      |
| TC-011 | Admin publishes a blog                           | 1. Login as Admin.<br>2. Select pending/draft blog.<br>3. Update publishing status.                                             | Blog status should be updated successfully.                                                      | Blog status updated to Published.                                                 | Pass      |
| TC-012 | Create category with authorized role             | 1. Login as Admin/Editor.<br>2. Submit category details.                                                                        | Category should be created successfully.                                                         | Category created successfully.                                                    | Pass      |
| TC-013 | Reader attempts category creation                | 1. Login as Reader.<br>2. Submit category creation request.                                                                     | Request should be rejected.                                                                      | Request rejected with 403 Forbidden.                                              | Pass      |
| TC-014 | Add comment to published blog                    | 1. Login as authenticated user.<br>2. Select published blog.<br>3. Submit comment.                                              | Comment should be created successfully.                                                          | Comment created successfully.                                                     | Pass      |
| TC-015 | Comment moderation                               | 1. Login as Admin/Editor.<br>2. Open a comment.<br>3. Change moderation status.                                                 | Comment status should be updated successfully.                                                   | Comment status updated successfully.                                              | Pass      |
| TC-016 | Generate AI blog                                 | 1. Login as authorized user.<br>2. Send topic to /api/ai/generate-blog.<br>3. Submit request.                                   | Gemini should return generated blog content.                                                     | Gemini returned generated blog content successfully.                              | Pass      |
| TC-017 | Summarize blog content                           | 1. Send blog content to /api/ai/summarize.<br>2. Submit request.                                                                | AI-generated summary should be returned.                                                         | Summary generated and returned successfully.                                      | Pass      |
| TC-018 | Keyword search                                   | 1. Send /api/search?q= request.<br>2. Enter a relevant keyword.                                                                 | Matching blog results should be returned.                                                        | Matching blogs returned via Atlas Search.                                         | Pass      |
| TC-019 | Semantic search                                  | 1. Send /api/search/semantic?q= request.<br>2. Enter a natural-language query.                                                  | Conceptually relevant blog results should be returned.                                           | Relevant blogs returned via Atlas Vector Search.                                  | Pass      |
| TC-020 | Invalid blog ID                                  | 1. Send blog request with invalid ID.<br>2. Submit request.                                                                     | System should return a validation/not-found error.                                               | 404/validation error returned as expected.                                        | Pass      |
| TC-021 | Invalid AI request                               | 1. Send AI request without required input.<br>2. Submit request.                                                                | System should return a validation error.                                                         | Validation error returned as expected.                                            | Pass      |
| TC-022 | Unauthorized role operation                      | 1. Login with restricted role.<br>2. Attempt an admin-only operation.                                                           | Request should be rejected with an authorization error.                                          | Request rejected with 403 Forbidden.                                              | Pass      |
| TC-023 | API rate limiting on repeated requests           | 1. Send repeated requests to the same endpoint in rapid succession, exceeding the configured limit.<br>2. Observe response.     | System should throttle excess requests and return a rate-limit error.                            | Requests beyond the threshold were rejected with 429 Too Many Requests.           | Pass      |
| TC-024 | Input sanitization on blog content (XSS attempt) | 1. Login as Author.<br>2. Submit a blog post containing an embedded script tag / malicious HTML.<br>3. Retrieve the saved blog. | Script/HTML content should be sanitized or rejected; no raw script should be stored or returned. | Malicious markup was stripped/escaped before storage; retrieved content was safe. | Pass      |
| TC-025 | Content discovery by category                    | 1. Send GET /api/blogs?category= request.<br>2. Provide a valid category filter.                                                | Only blogs belonging to the specified category should be returned.                               | Filtered results correctly returned blogs matching the category.                  | Pass      |

## 5. Test Case Summary

The testing covers:

### Positive Scenarios

- Successful registration
- Successful login
- Successful JWT authentication
- Successful blog creation
- Successful blog retrieval
- Successful blog update
- Successful category creation
- Successful comment creation
- Successful comment moderation
- Successful AI generation
- Successful AI summarization
- Successful keyword search
- Successful semantic search
- Successful category-based content discovery

### Negative Scenarios

- Duplicate registration
- Invalid login
- Missing JWT
- Unauthorized blog creation
- Unauthorized blog modification
- Unauthorized category management
- Invalid IDs
- Invalid AI requests
- Restricted administrative operations
- Excessive requests (rate limiting)
- Malicious input / XSS attempt

## 6. Bug Tracking

No defects were identified during UAT.

## 7. UAT Result Summary

|                  |    |
|------------------|----|
| Total Test Cases | 25 |
| Passed           | 25 |
| Failed           | 0  |
| Blocked          | 0  |

OVERALL UAT STATUS: PASSED

## 8. Sign-Off

|                         |                   |
|-------------------------|-------------------|
| Tester Name             | R. Hariharan      |
| Date of Test Completion | 26 September 2026 |
| Signature               |                   |